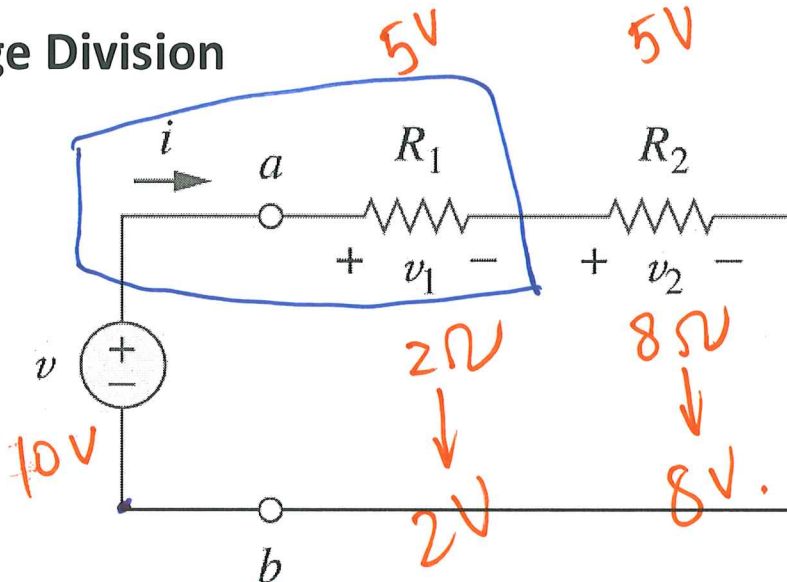


L#05: Voltage and Current Division

2018-02-13

Voltage Division



Series:

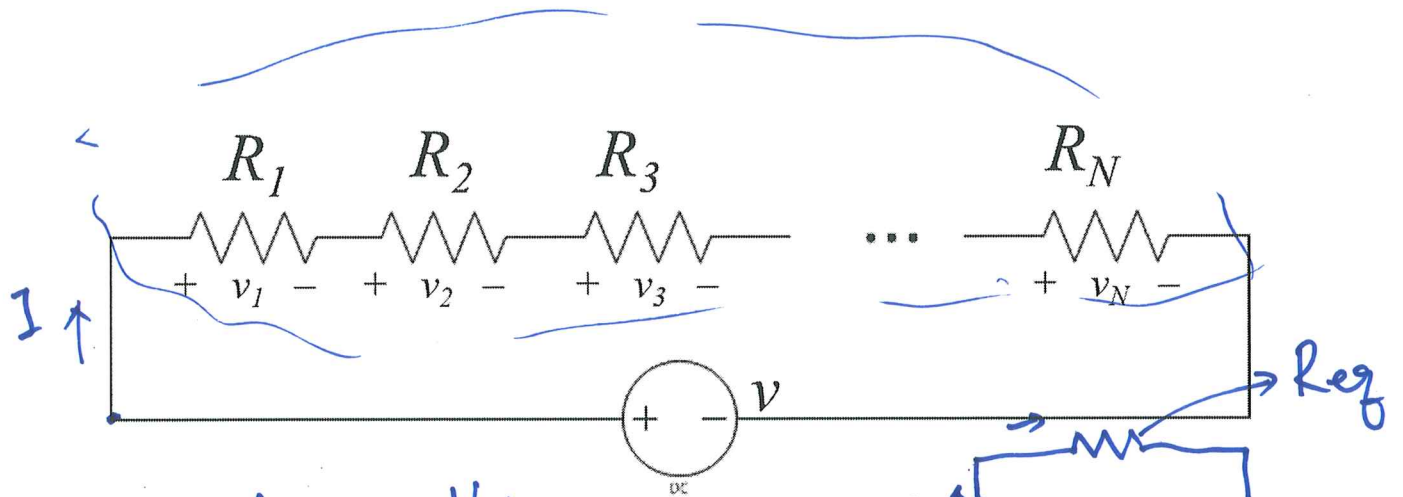
Voltage division:

$$-V + v_1 + v_2 = 0 \Rightarrow \underline{V = v_1 + v_2}$$

$$V = iR_1 + iR_2 \Rightarrow i = \frac{V}{R_1 + R_2}$$

$$\boxed{v_1} = i \cdot R_1 = \frac{V}{R_1 + R_2} \cdot R_1 = \boxed{V \cdot \frac{R_1}{R_1 + R_2}}$$

$$v_2 = iR_2 = \frac{V}{R_1 + R_2} \cdot R_2 = V \cdot \frac{R_2}{R_1 + R_2}$$

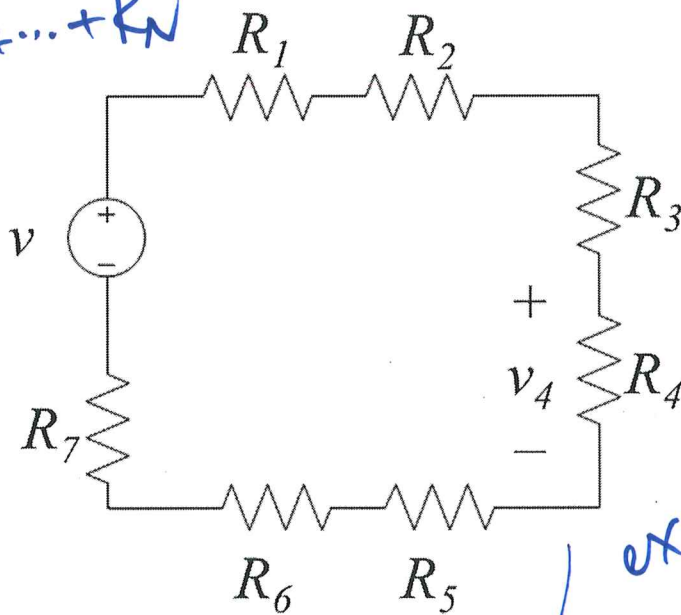


$$v = v_1 + v_2 + \dots + v_N$$

$$I R_{eq} = I R_1 + I R_2 + \dots + I R_N$$

$$R_{eq} = R_1 + R_2 + \dots + R_N$$

$$= \sum_{i=1}^N R_i$$



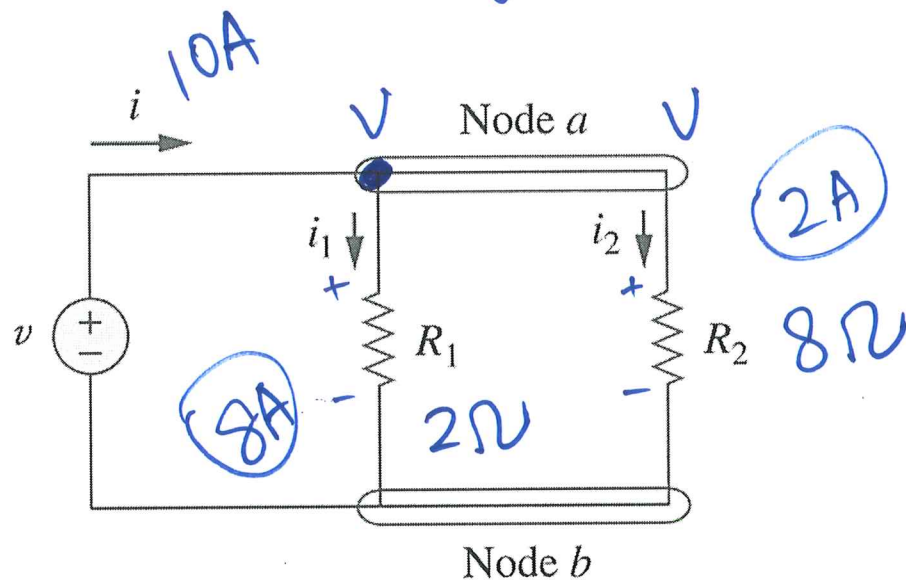
expanded voltage division.

$$v_4 = \frac{R_4}{\sum_{i=1}^7 R_i} v$$

Current division:

Parallel:

$$i + (-i_1 - i_2) = 0.$$



$$i = i_1 + i_2 = \frac{V}{R_1} + \frac{V}{R_2} = V \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$= V \cdot \frac{R_1 + R_2}{R_1 R_2} \Rightarrow$$

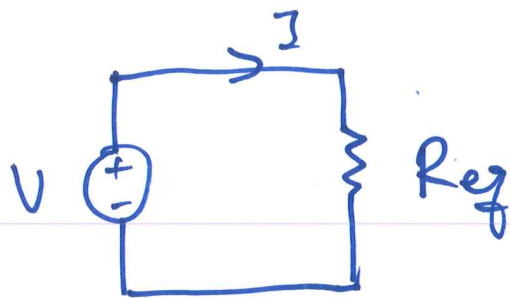
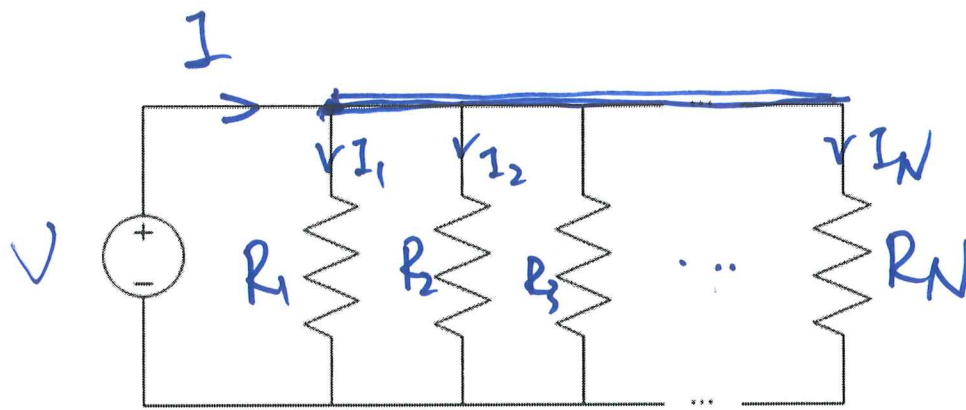
$$\frac{R_1 R_2}{R_1 + R_2} \times i = V \cdot \frac{R_1 + R_2}{R_1 R_2} \Rightarrow V = i \cdot \frac{R_1 R_2}{R_1 + R_2}$$

$$\Rightarrow i_1 = \frac{V}{R_1} = i \cdot \frac{R_1 R_2}{R_1 + R_2} \cdot \frac{1}{R_1}$$

For two resistors:

$$i_1 = i \cdot \frac{R_2}{R_1 + R_2}$$

$$i_2 = i \cdot \frac{R_1}{R_1 + R_2}$$



$$I = I_1 + I_2 + \dots + I_N$$

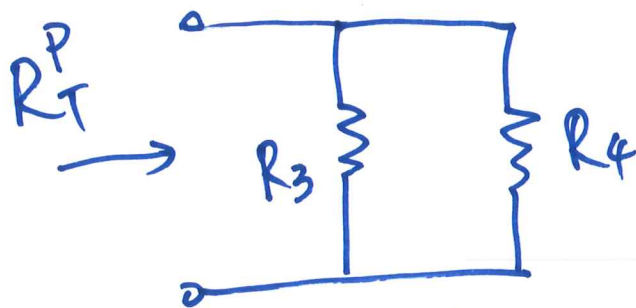
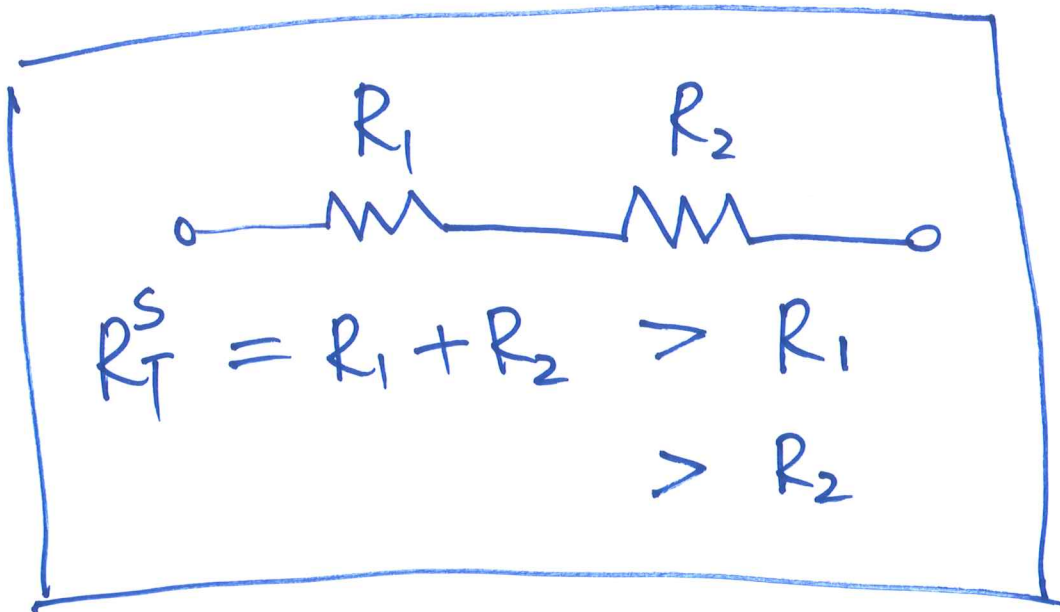
$$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} + \dots + \frac{V}{R_N}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}$$

$$\frac{1}{R_{eq}} = \sum_{i=1}^N \frac{1}{R_i}$$

$$\begin{aligned} \frac{1}{R_T} &= \frac{1}{R_1} + \frac{1}{R_2} \\ &= \frac{R_1 + R_2}{R_1 R_2} \end{aligned}$$

$$\Rightarrow R_T = \frac{R_1 R_2}{R_1 + R_2}$$



$$R = \rho \frac{l}{A}$$

$$R_T^P = \frac{R_3 R_4}{R_3 + R_4} < R_3$$

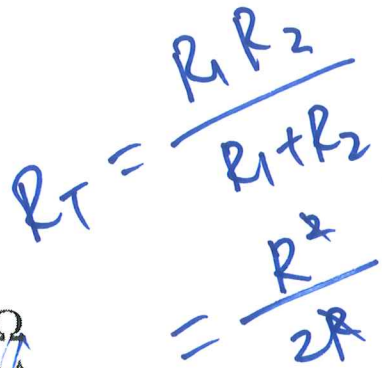
$$< R_4$$

8

6



Determine R_{eq} .



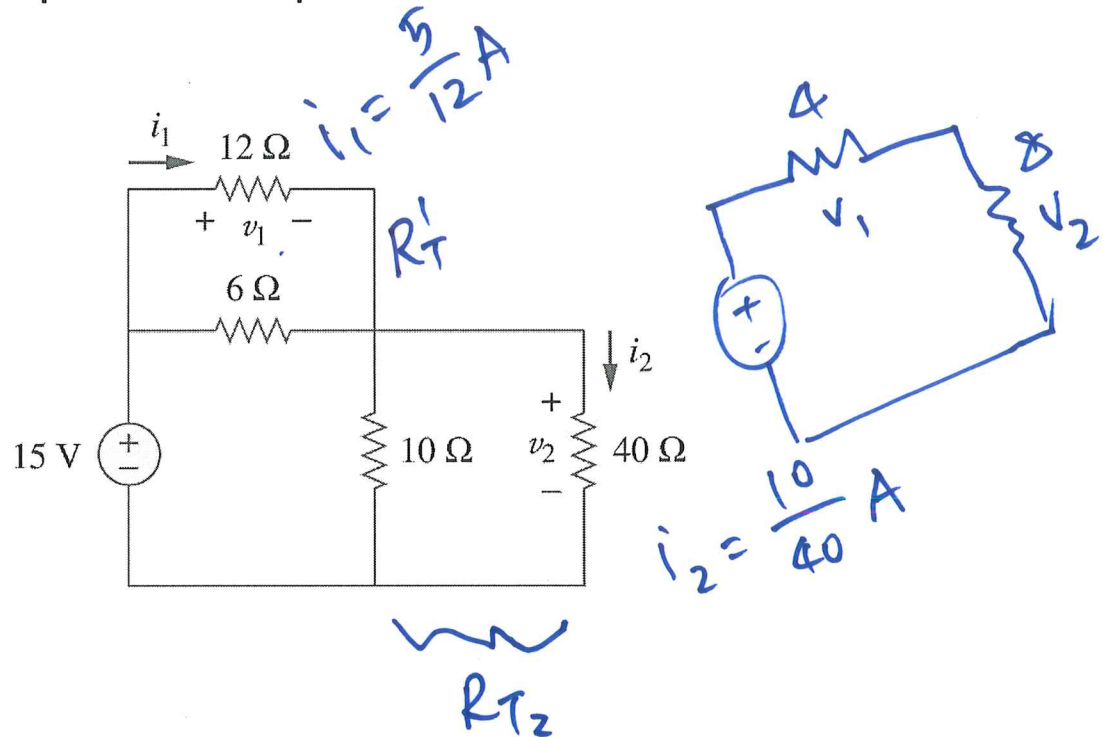
② $12 \parallel 4 = \frac{4 \times 12}{4 + 12} = 3$

③ $6 \parallel 6 = 3 \Omega$

④ $R_{eq} = 6 \Omega$

Example #2:

Find v_1 and v_2 in the circuit. Also calculate i_1 and i_2 and the power dissipated in the $12\ \Omega$ and $40\ \Omega$ resistors.



$$R_{T1} = \frac{12 \times 6}{12 + 6} = 4; \quad R_{T2} = \frac{10 \times 40}{10 + 40} = 8$$

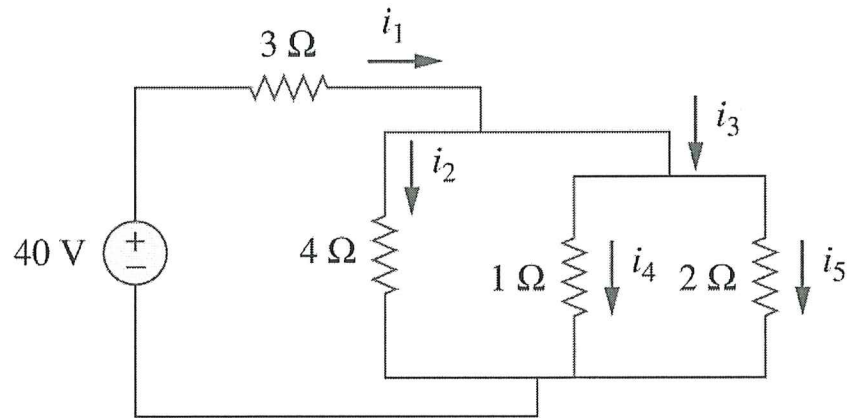
$$V_1 = 15 \times \frac{4}{4 + 8} = 5 \text{ V}, \quad V_2 = 10 \text{ V}.$$

$$P_{12} = \frac{V_1^2}{12} = \frac{25}{12}; \quad P_{40} = \frac{V_2^2}{40} = \frac{100}{40} = 2.5 \text{ W}.$$

Example #3:

Determine i_1 to i_5 .

homework



Q1: Calculate the amount of charge represented by four million protons.

$$e = -1.602 \times 10^{-19} \text{ C}$$

$$Q = 4 \times 10^6 \times 1.602 \times 10^{-19} \text{ C}$$

Q2: The total charge entering a terminal is given by $q = (10 - 10e^{-2t})$ mC. Find the **current** at $t = 0.5$ s.

$$q(t) = 10 - 10e^{-2t} \text{ mC}$$

$$i(t) = \frac{dq(t)}{dt} = 0 - (10e^{-2t} \cdot -2)$$

$$i(t) = 20e^{-2t} \text{ mA}$$

$$i(0.5) = 20 \boxed{e^{-1}} \text{ mA}$$

Q3: Determine the total charge entering a terminal between $t = 1$ s and $t = 2$ s if the current passing the terminal is:

$$\underline{i = (3t^2 - t) \text{ A}}$$

$$i = \frac{dq}{dt}$$

$$q = \int_{t_1}^{t_2} i(t) dt$$

$$= \int_1^2 (3t^2 - t) dt$$

$$= \left(t^3 - \frac{1}{2} t^2 \right) \Big|_1^2$$

$$= \left(2^3 - \frac{1}{2} \times 2^2 \right) - \left(1^3 - \frac{1}{2} 1^2 \right)$$

Example #1:

To move charge $q = 2\text{C}$ from point a to b requires -30 J . Find the voltage drop v_{ab} .

$$\cancel{V_{ab} = \frac{Q}{W} = .}$$

$$V_{ab} = \frac{W}{Q} = \frac{-30\text{ J}}{2\text{ C}} \\ = -15\text{ V.}$$

Example #2:

Find the power delivered to an element at $t = 5 \text{ ms}$ if the voltage is $v = 2i \text{ V}$ the current entering its positive terminal is: $i = 5 \cos 60\pi t \text{ A}$.

$$p(t) = v(t) \cdot i(t)$$

$$= 5 \cos 60\pi t \times 2 \times 5 \cos 60\pi t$$

$$= 50 \cos^2 60\pi t \text{ (W)}$$

$$\pi = 180^\circ$$

$$p(5\text{ms}) = 50 \cos^2 (60 \times 180^\circ \times 5 \times 10^{-3}) \text{ W}$$

$$(17.27 \text{ W})$$

Example #3:

A stove element draws 15 A when connected to a 240 V supply. How long does it take to consume 60 kJ?

Example #4:

Calculate the power supplied or absorbed by each element.

